

Paper Portait PCB - EN 62368-1 Safety Certification Documentation

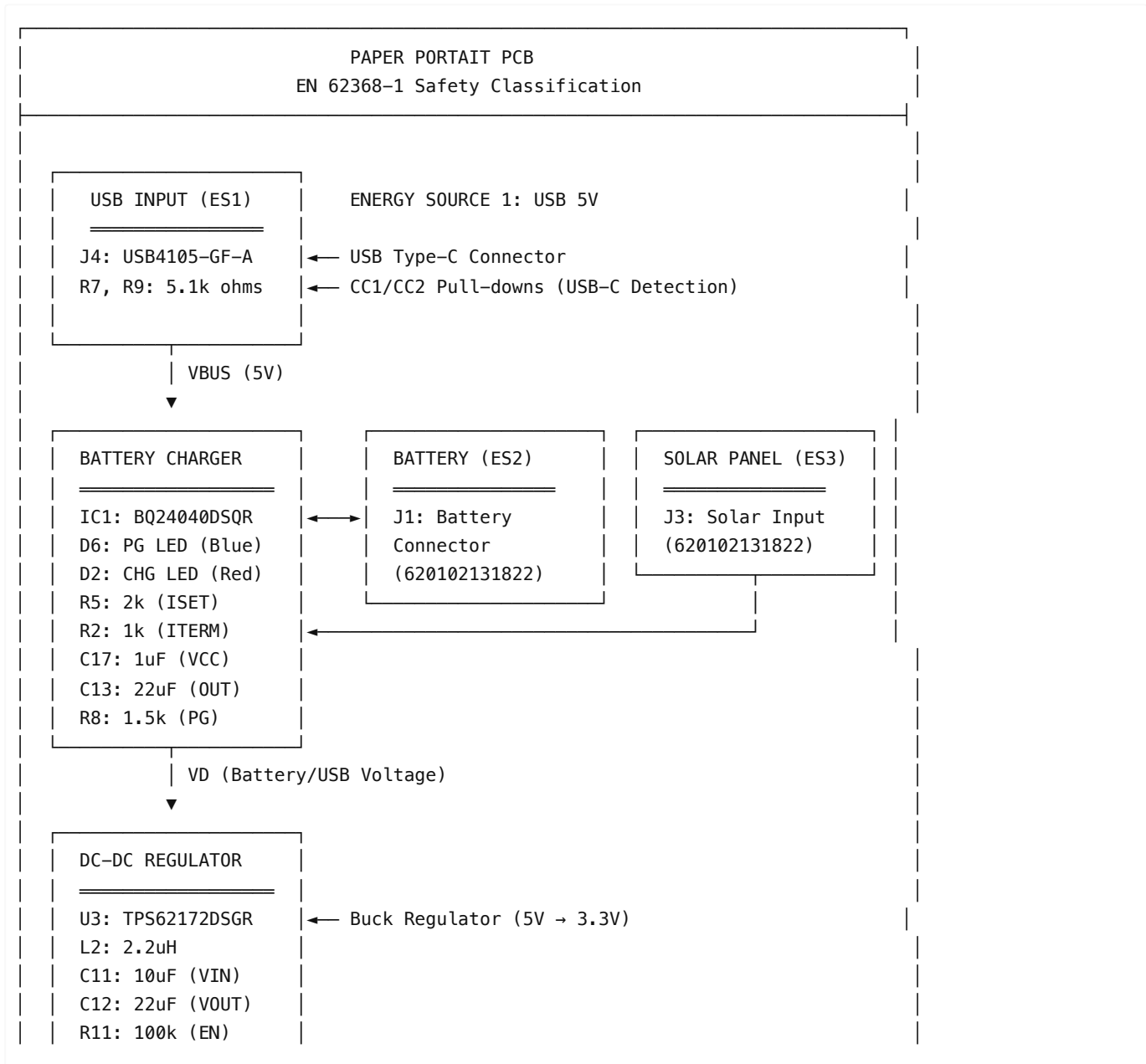
Document Version: 1.0 Date: 2026-02-11 Product: Paper Portait PCB Manufacturer: Wanyeki Technologies LLC

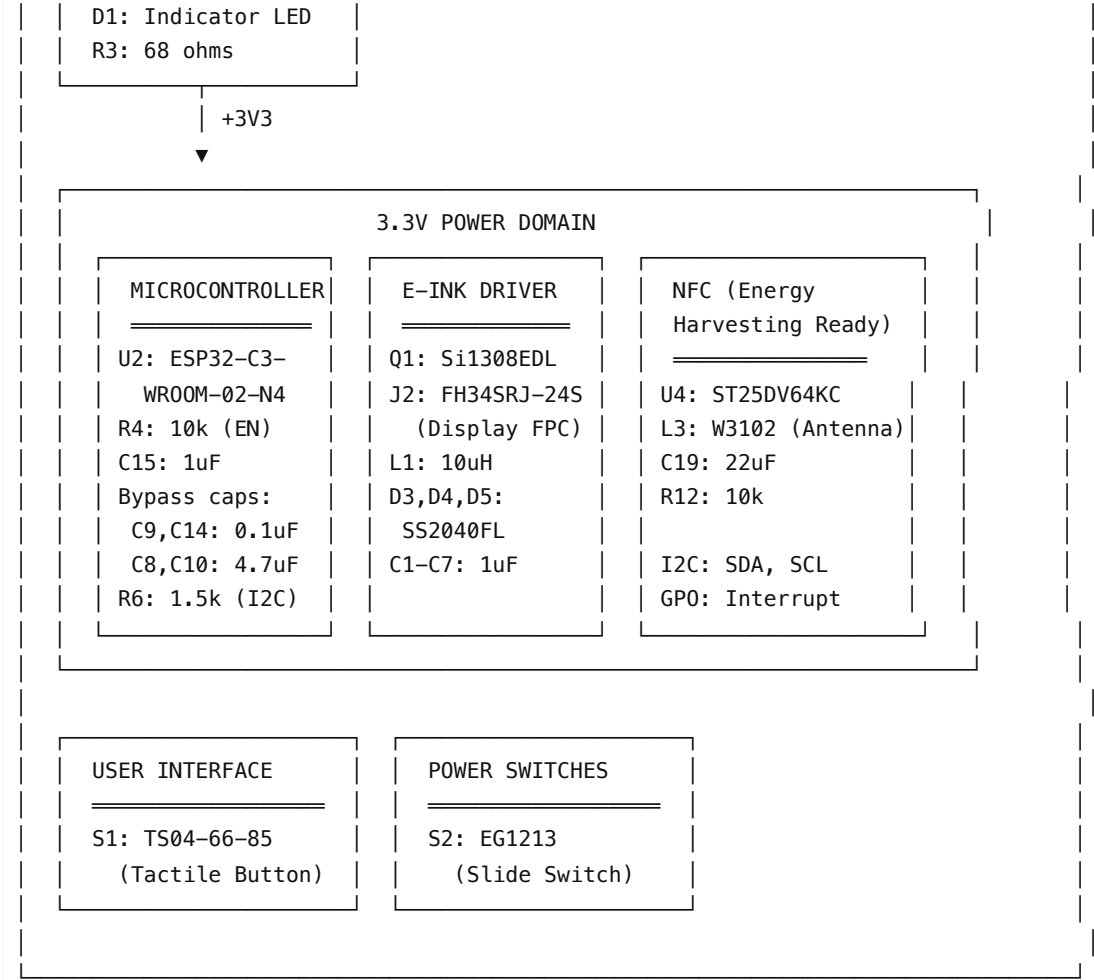
1. Product Overview

The Paper Portait PCB is a low-power IoT device featuring:

- ESP32-C3 WiFi/BLE microcontroller
- E-Ink display interface
- NFC communication with energy harvesting capability
- USB Type-C charging
- Li-ion battery power management

2. Safety Block Diagram





3. Energy Source Classification

Energy Source	Type	Voltage	Current Limit	Classification
ES1 - USB	External DC	5V	500mA - 1.5A	ES1 (≤60V DC)
ES2 - Battery	Internal DC	3.7V nominal	Protected cell	ES1 (≤60V DC)
ES3 - Solar Panel	External DC (J3)	4.2-6V	Panel dependent	ES1 (≤60V DC)
ES4 - NFC Harvesting	RF Induced	<3.3V	<10mA	ES1

All circuits operate within ES1 classification - No hazardous voltage levels present.

4. USB-Connected Components

These components are directly in the USB power/data path and require special attention for EN 62368-1 compliance:

Reference	Part Number	Manufacturer	Description	Safety Relevance
J4	USB4105-GF-A	GCT	USB Type-C Connector	Primary interface, ESD protection point
R7	ERJ-1GNF5101C	Panasonic	5.1k ohm (CC1)	USB-C configuration resistor
R9	ERJ-1GNF5101C	Panasonic	5.1k ohm (CC2)	USB-C configuration resistor

IC1	BQ24040DSQR	Texas Instruments	Battery Charger IC	Receives VBUS directly, thermal protection
C17	CL03A105MO3NRNH	Samsung	1uF/16V	Input filtering on USB side

USB Port Specifications

- **Connector Type:** USB Type-C (UFP - Upstream Facing Port)
- **Power Role:** Sink only (device charges from host)
- **Data Lines:** Not connected (power only)
- **ESD Rating:** Verify J4 (USB4105-GF-A) meets IEC 61000-4-2 Level 4

5. Complete Bill of Materials (BOM)

5.1 Power Input & Battery Charging

Ref	Qty	Part Number	Manufacturer	Value	Footprint	Safety Notes
J4	1	USB4105-GF-A	GCT	USB-C	GCT_USB4105-GF-A	USB INPUT - Primary safety interface
R7	1	ERJ-1GNF5101C	Panasonic	5.1k ohm	R_0201	USB CC1 pin
R9	1	ERJ-1GNF5101C	Panasonic	5.1k ohm	R_0201	USB CC2 pin
IC1	1	BQ24040DSQR	Texas Instruments	-	SON40P200X200X80-11N	Charger IC - thermal protection built-in
J1	1	620102131822	Wurth Elektronik	-	620102131822	Battery connector
J3	1	620102131822	Wurth Elektronik	-	620102131822	Solar panel charger input
D2	1	150060RS75000	Wurth Elektronik	Red LED	LED_0603	CHG indicator
D6	1	150060BS75000	Wurth Elektronik	Blue LED	LED_0603	Power Good indicator
R2	1	ERJ-1GNF1001C	Panasonic	1k ohm	R_0201	ITERM setting
R5	1	ERA-1AEB202C	Panasonic	2k ohm	R_0201	ISET (charge current)
R8	1	ERJ-1GNF1501C	Panasonic	1.5k ohm	R_0201	PG pullup
C17	1	CL03A105MO3NRNH	Samsung	1uF/16V	C_0201	VCC decoupling
C13	1	CL10A226MP8NUNE	Samsung	22uF/10V	C_0603	Output capacitor

5.2 Voltage Regulation (3.3V Supply)

Ref	Qty	Part Number	Manufacturer	Value	Footprint	Safety Notes
U3	1	TPS62172DSGR	Texas Instruments	-	SON50P200X200X80-9N	Buck regulator, thermal shutdown

L2	1	MLZ1608N2R2LT000	TDK	2.2uH	L_0603	Power inductor
C11	1	CL10A106MP8NNNC	Samsung	10uF/10V	C_0603	Input capacitor
C12	1	CL10A226MP8NUNE	Samsung	22uF/10V	C_0603	Output capacitor
R11	1	ERJ-1GNF1003C	Panasonic	100k ohm	R_0201	Enable pullup
D1	1	150060RS75000	Würth Elektronik	Red LED	LED_0603	Power indicator
R3	1	ERJ-1GNF68R0C	Panasonic	68 ohm	R_0201	LED current limit

5.3 Microcontroller

Ref	Qty	Part Number	Manufacturer	Value	Footprint	Safety Notes
U2	1	ESP32-C3-WROOM-02-N4	Espressif	-	ESP32-C3-WROOM-02	WiFi/BLE module, FCC/CE certified
R4	1	ERJ-1GNF1002C	Panasonic	10k ohm	R_0201	EN pullup
R6	1	ERJ-1GNF1501C	Panasonic	1.5k ohm	R_0201	I2C pullup
C15	1	CL03A105MO3NRNH	Samsung	1uF/16V	C_0201	Decoupling
C8	1	CL05A475MP5NRNC	Samsung	4.7uF/10V	C_0402	Bypass
C10	1	CL05A475MP5NRNC	Samsung	4.7uF/10V	C_0402	Bypass
C9	1	CL03A104KP3NNNC	Samsung	0.1uF/10V	C_0201	Bypass
C14	1	CL03A104KP3NNNC	Samsung	0.1uF/10V	C_0201	Bypass

5.4 E-Ink Display Driver

Ref	Qty	Part Number	Manufacturer	Value	Footprint	Safety Notes
Q1	1	SI1308EDL-T1-GE3	Vishay	N-MOSFET	SOT-323	Display power switch
J2	1	FH34SRJ-24S-0.5SH(99)	Hirose	24-pin FPC	HRS_FH34SRJ-24S	Display connector
L1	1	MLZ1608N100LT000	TDK	10uH	L_0603	Filter inductor
D3	1	SS2040FL-AU_R1_000A1	Panjit	Schottky	SODFL3618X108N	Protection diode
D4	1	SS2040FL-AU_R1_000A1	Panjit	Schottky	SODFL3618X108N	Protection diode
D5	1	SS2040FL-AU_R1_000A1	Panjit	Schottky	SODFL3618X108N	Protection diode
R1	1	KDV02FR470ET	Ohmite	0.47 ohm	R_0201	Current sense
C1	1	CL03A105MO3NRNH	Samsung	1uF/16V	C_0201	Decoupling
C2	1	CL03A105MO3NRNH	Samsung	1uF/16V	C_0201	Decoupling
C3	1	CL03A105MO3NRNH	Samsung	1uF/16V	C_0201	Decoupling
C4	1	CL03A105MO3NRNH	Samsung	1uF/16V	C_0201	Decoupling
C5	1	CL03A105MO3NRNH	Samsung	1uF/16V	C_0201	Decoupling

C6	1	CL03A105MO3NRNH	Samsung	1uF/16V	C_0201	Decoupling
C7	1	CL03A105MO3NRNH	Samsung	1uF/16V	C_0201	Decoupling

5.5 NFC Subsystem

Ref	Qty	Part Number	Manufacturer	Value	Footprint	Safety Notes
U4	1	ST25DV64KC-IE8T3	STMicroelectronics	-	TSSOP8_STM	NFC/RFID with energy harvesting
L3	1	W3102	Pulse (Yageo)	-	IND_W3102	NFC antenna coil
C19	1	CL10A226MP8NUNE	Samsung	22uF/10V	C_0603	VCC_NFC decoupling
R12	1	ERJ-1GNF1002C	Panasonic	10k ohm	R_0201	GPO pullup

5.6 User Interface & Switches

Ref	Qty	Part Number	Manufacturer	Value	Footprint	Safety Notes
S1	1	TS04-66-85-BK-160-SMT	CUI Devices	Tactile	SW_TS04-66-85	User button
S2	1	EG1213	E-Switch	SPDT	SW_EG1213	Power slide switch

6. EN 62368-1 Safety Compliance Summary

6.1 Applicable Clauses

Clause	Requirement	Compliance Status
4.1	Electrical energy source classification	ES1 - All circuits ≤60V DC
5.4	Limited Power Source (LPS)	USB input is LPS from host
6.4	Thermal requirements	IC1, U3 have thermal protection
9.2	Mechanical hazards	No sharp edges, proper enclosure required
Annex Q	Battery requirements	Single cell Li-ion, protected

6.2 Safety Features

- Overcurrent Protection:** BQ24040 includes programmable charge current limiting
- Thermal Protection:** BQ24040 reduces charge current at high temperatures
- Overvoltage Protection:** TPS62172 operates within safe input range
- ESD Protection:** USB connector rated for ESD per IEC 61000-4-2
- Low Voltage Operation:** All circuits operate at ≤5V (ES1 classification)

7. Regulatory Certifications (Module Level)

Component	Certification	Notes
ESP32-C3-WROOM-02	FCC, CE, IC, MIC, KCC, NCC	Pre-certified WiFi/BLE module
ST25DV64KC	CE	NFC Forum Type 5 Tag

8. Document Revision History

Version	Date	Author	Changes
1.0	2026-02-11	-	Initial release

This document is intended to support EN 62368-1 safety certification. Final compliance determination must be made by an accredited test laboratory.